



Teachers' attitudes towards inclusive education of children with special educational needs and disabilities in central Asia

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ARTICLE INFO

Keywords:

Central Asia
Children with Disabilities
Children with Special Educational Needs and Disabilities
Inclusive Education
Inclusion
Teachers' Attitudes

ABSTRACT

Teachers' attitudes towards inclusive education, particularly towards students with special educational needs and disability (SEND), have become a focus of research in the past years. In many countries, the process of the implementation of inclusive education has either not been smooth or, in some cases, stalled by negative attitudes of teachers. Teachers are regarded as key drivers of the implementation of inclusive education as they are at schools and classrooms, which are strategic delivery points. Central Asian countries have embarked on a process of implementing inclusive education in their respective education systems. Therefore, in this quantitative study, using regression analysis, and one-way ANOVA, the attitudes of eight hundred and sixty-nine (869) teachers from Central Asian countries i.e. Kazakhstan, Kyrgyzstan, Tajikistan, and Uzbekistan were surveyed. The analysis shows a noticeable shift from the segregation strategy embedded in defectology to inclusion and equity for students with SEND. However, the teachers' attitudes still vary, the teachers could be classified into six groups according to their settings towards inclusive education.

1. Introduction

Teachers' attitudes are regarded as central towards the success of inclusive education of children with special needs and disabilities.¹ Inclusive education provides an opportunity for students with different backgrounds, characteristics, and abilities to be educated in the same less restrictive educational environment (Makoelle, 2016). Starting with the adoption of the Salamanca Statement in 1994, many countries have decided to change their schooling system to embrace inclusive education. This was done by including students with learning difficulties, special educational needs, and disabilities in mainstream education. This transition from exclusive segregated schooling to a more inclusive model has been met with doubts and, in some cases, resistance. Teachers are expected to embark on a paradigm shift in terms of their teaching and learning practices. However, change is not an easy process, and thus teacher beliefs, conceptions, and attitudes towards special needs and

disabilities can either become a barrier or an enabler of the inclusion process.

This study analyzed the attitudes of teachers towards inclusive education of students with special educational needs and disabilities (SEND) in four central Asian countries, Kazakhstan, Kyrgyzstan, Tajikistan, and Uzbekistan are the four post-socialist countries where inclusive education is becoming both a goal and a challenge for social policy.² On the one hand, these countries are rather similar: they are united by a common Soviet legacy and largely inherited the Soviet system of social policy. On the other hand, countries differ greatly in socio-economic and cultural characteristics. Responding to global and local challenges, the four post-socialist countries of Central Asia began to search for ways of developing inclusion in education. At the beginning of the article based on the literature review, we identify the key initiatives, barriers, and drivers of inclusive education in these countries to contextualize socio-cultural, organizational, and legislation

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¹ In this paper we use concept 'Children with disabilities' identifying "disability results from the interaction between persons with impairments and attitudinal and environmental barriers that hinders their full and effective participation in society on an equal basis with others" (Convention on the Rights of Persons with Disabilities, <https://www.ohchr.org/en/instruments-mechanisms/instruments/convention-rights-persons-disabilities>). Concept of 'Children with special educational needs and disability' (SEND) arises from wording in the questionnaire and used in this paper for the analysis and interpretation of the survey results,

² Turkmenistan, as one more post-socialist country of Central Asia, was excluded from the analysis because of the lack of comparable empirical data.

circumstances affecting teacher's attitudes formation. Next, in this study based on teacher survey data addresses the following overarching research question: what are the teachers' attitudes towards inclusive education of children with special educational needs and disabilities in Kazakhstan, Kyrgyzstan, Tajikistan, and Uzbekistan? Answering this question we contribute to the discussion on the prospects for inclusive education in the countries of Central Asia, since the attitudes of teachers, as shown below, are crucial in this matter.

2. Literature review

2.1. The concepts of inclusive education, SEND, and teachers' attitudes

Inclusive education is regarded as an elusive concept. However, its understanding is central to broadening the participation of students in the teaching and learning process regardless of their differences, which may include such barriers as disability, socio-economic status, gender, culture, language, ethnicity, race, and health (Makoelle, 2014a). While the special needs approach to teaching and learning was about making students fit into the normative educational process, inclusive education is about changing the teaching and learning approach so that education can include and respond to each student's needs. The transition towards inclusive education often requires a change of teaching and learning practices but, most importantly, the changing of teacher's beliefs, values, attitudes, and knowledge about teaching and learning.

SEND represents a diverse group of students, including learning difficulties, behavior difficulties, autism spectrum disorder, and others (Mamas, Daly, & Schaeffer, 2016), who need a special educational environment to develop social skills and academic performance (Fernandez-Villardón, Pilar Alvarez, & Itxaso, 2020).

Haddock and Maio (2012:114) define an attitude as a psychological tendency that is expressed by evaluating a particular entity with some degree of favor or disfavor. It is an evaluative process of making a judgment about liking or disliking something or someone. Therefore, the notion of teacher attitude may be understood to mean the way the teacher perceives and thinks, which may lead to how they will act in their teaching practices. This means their choices to like or dislike certain aspects of the professional engagement. According to Nel et al. (2011), the notion of teachers' attitudes cannot be divorced from the interaction with context, others, and the situation. Teachers' attitudes are not a static concept but are also based on teachers' constructions of what they perceive as favorable or unfavorable which, in turn, is derived from wider social and cultural context.

2.2. Teachers' attitudes towards inclusion: An international perspective

Measuring the attitudes of teachers toward inclusive education is a complex process as the concept of teacher attitude is understood differently by different teachers in different contexts. However, according to Ewing, Monsen, and Kielblock (2018), several teachers' attitudes measuring scales have been used, i.e., the Multidimensional Attitude Towards Inclusive Education Scale (MATIES), Sentiments, Attitude and Concerns about Inclusive Education Revised Scale (SACIER-R), and Teacher's Attitudes towards Inclusion Scale (TAIS). The purpose of determining teachers' attitudes towards inclusive education is to understand what causes them and how they could be improved. Gyimah et al. (2010) posit that the attitudes of teachers are critical for the delivery of the quality of education and the interaction of teachers and students. They go further to suggest that teachers' attitudes also differ greatly according to different educational fields. However, while factors influencing teachers' attitudes revolve around the same factors, the influence of teachers' attitudes toward inclusive education is varied and the literature seems to suggest that different factors dominate teachers' attitudes in different contexts, situations, and countries. There is also a dichotomy between rural and urban contexts.

Several factors are known to influence the attitudes of teachers

towards inclusive education internationally. Ratification of the UN Convention on the Rights of People with Disabilities is among international factors that may influence teachers' attitudes towards school inclusion positively as they may be expected to adopt it in their teaching practices (Baldiga et al., 2017). The ratification promotes the development of inclusive policy and practice in the country, including changes in legislation, training programs for teachers, etc. Agavelyan, Aubakirova, Zhomartova, and Burdina (2020) have identified the training of teachers in preparation for inclusive teaching as pivotal in improving the attitudes of teachers towards inclusive education in Kazakhstan. The issue of training is also prominent in many studies conducted, especially in developing countries. For instance, in Ghana, according to the study conducted by Gyimah et al. (2010), the factors that influenced teachers' attitudes towards inclusive education included understanding of inclusive education, type of disability, gender, teaching experience, teacher beliefs, and available support. Remarkably, in this context, it seemed as though teacher training was the most cited factor that influenced the attitudes of teachers negatively. While teacher training seems influential towards the attitude of teachers, Singh, Kumar, and Singh (2020) caution that pre-service teachers tend to have more positive attitudes toward inclusive education than their in-service counterparts. At the same time, Unianu (2012) found that the more experienced teachers were the ones who did not struggle much when it came to embracing the notion of inclusive teaching. Long professional experience is associated with self-confidence and dealing with students from diverse ethnic and cultural backgrounds. This is also confirmed by Bandyopadhyay and Dhara (2021). Therefore, the role of the teacher in inclusive education has a bearing on their attitudes towards inclusion. For instance, according to Makoelle (2014b), mainstream teachers often regard dealing with students with special educational needs and disability as an added extra-load to their teaching responsibilities.

The character and nature of a teacher are also important in determining their attitude. For instance, Bandyopadhyay and Dhara (2021) found that the teacher's level of education, self-efficacy, and gender may impact their attitudes toward inclusive education. They argue that teachers with a higher level of education are likely to show a positive attitude towards inclusive education. In their assertion, male teachers tend to show a positive attitude towards inclusive education. However, the issue of gender is contested by other studies as a determining factor in the attitudes towards inclusion. For instance, Saloviita (2020) states that female teachers are more positive towards school inclusion, but it is not the only factor. The attitudes of teachers could also depend on the socio-cultural background (social and cultural beliefs) and teacher socialization (training and development) in different education systems. Furthermore, it seems as though self-efficacy and teacher work orientation were less significant to teacher's attitudes in developed countries than in developing countries. In his work about changing teacher beliefs and attitudes towards inclusive education, Makoelle (2014b) avers that the teacher-driven process of change is critical for inclusion to thrive. Makoelle further postulates that for teachers to change their views about inclusion, the establishment of professional learning communities, also called communities of practice or inquiry is very important, as they offer the teachers the platform to reflect, discuss, and critique their practices, including their attitudes towards inclusive education process. This view is echoed by San Martín, Ramírez, Calvo, Muñoz-Martínez, and Sharma (2021), who contend that collaboration and working with others were influential factors in changing teachers' attitudes towards inclusive education.

Teachers' attitudes towards inclusive education of children with disabilities are connected with national sociocultural backgrounds and legislation. To understand the nature as well as similarities and differences in teachers' attitudes, we identify the key incentives, drivers, and barriers towards school inclusion in four countries of Central Asia.

2.3. Inclusive education in Central Asia: Drivers and barriers

In the first half of the 1990s, in all four countries of Central Asia, social policy was based on the medical model of disability, which is reflected in the segregation of children with disabilities in society. This system of segregation scientifically referred to in the Soviet Union as defectology assumes that students can be better taught in separate settings to remediate their special needs. The placement is carried out by a special commission once a special need or disability has been identified. The defectologists who are specialists in this field are charged with the responsibility of providing specialized educational support (Makoelle, 2020). In particular, this policy was reflected in the education of children with disabilities in specialized correctional, often closed schools (Akkozhoeva, 2020; Makoelle, 2020; Sagyndyk & Omorox, 2021). The direction towards inclusive education started in four countries of Central Asia at different times. Kazakhstan is a pioneer in the region, which earlier adopted the first legislative acts on inclusive education, and ratified the Convention on the Rights of Persons with Disabilities (hereinafter – Convention) in 2015,³ Kyrgyzstan⁴ and Uzbekistan⁵ have ratified the Convention quite recently – in 2019 and 2021 accordingly, Tajikistan is only setting itself such a task.

Nevertheless, certain incentives towards school inclusion, supported by international and local grassroots organizations, are observed today throughout the region. Starting from the 2010s, all four countries of Central Asia began to develop and adopt legislative acts that create legal, infrastructural, and organizational opportunities for inclusive education and started to implement Roadmaps to achieve significant indicators of secondary school inclusion.

Despite several important initiatives in promoting inclusive education, there are some barriers to inclusion universal for the region and unique for each country. There are common problems connected with the Soviet legacy i.e., rigid adherence to educational support based on the medical model. This is reflected in the following way: according to UNICEF and local statistical agencies, the proportion of children with disabilities in the total child population in the 2010s varies between 0, 8 and 3 % in the region as a whole (UNICEF, 2014; UNICEF, 2017).⁶ According to the World Health Organization, it is seriously lower than the global average of 10–15 % (UNICEF, 2017). At the same time, the case of Kazakhstan shows that after the ratification of the Convention, the registration of children as disabled began to grow (Turlubekova and Bugubayeva, 2021). According to UNICEF (2017), ratification of the Convention becomes a driver for the development of social policy measures for children with disabilities, aimed at reducing barriers to obtaining disability status, such as bureaucratic obstacles in applying for special benefits, as well as discrimination and stigma towards people with disability at large. In Tajikistan, the situation with the assignment of disability status is complicated by the lack of uniform standards and requirements within the country, when Pedagogical, Medical, Psychological Consultation, and Medical, and Pedagogical Commissions as governmental bodies responsible for diagnosing disability as well as determine the individual educational route of every child use different criteria and tools for determining disability. Financial expenses for

proving a child's disability also is a barrier in Tajikistan.⁷ The medicalization of disability could be observed in Uzbekistan where the disciplines on inclusive education are taught at the Faculty of defectology, where the education is based on the deficit model (United Nations, 2019).

Be that as it may, the question remains: to what extent are secondary schools in the four countries of Central Asia inclusive? It is difficult to answer this question due to factors such as the lack of statistical data, primarily international and comparative, the number and dynamics of inclusive schools and classes, the proportion of children with disabilities studying in different types of schools, etc. Officially published statistics are inconsistent and based on different methodologies, this is why any comparisons are difficult or even impossible. Four countries of Central Asia use different terminology and indicators, to describe school inclusion: schools ready for inclusion, schools created condition for inclusion⁸ (Kazakhstan), mainstream educational institutions for a child with special needs⁹ (Uzbekistan), organizations (mainstream schools) for a child with special needs¹⁰ (Kyrgyzstan). All these terms are not properly operationalized in legislative acts.

As for Tajikistan, discrepancies in the statistical data are observed. According to different sources from 26 % (UNESCO, 2021) to 46 % of children with disabilities (Ruziev, Abdurahimov, Gaibov, & Bogodirov, 2019) study in mainstream schools. The further development of school inclusion in Tajikistan is constrained by the lack of educational standards, the medicalization of approaches to disability, and the lack of competencies among teachers (UNESCO, 2021). Despite the developed educational standards and training programs for teachers in Kyrgyzstan, the proportion of children with disabilities who do not attend school is twice as high as among those who attend school. According to Sagyndyk and Omorox, 13, 6 % of children with disabilities study in mainstream schools in 2021 (Sagyndyk and Omorox, 2021). One of the main reasons is poor infrastructure because only 8 % of secondary schools in Kyrgyzstan by 2020 had the infrastructure to teach children with disabilities (UNESCO, 2021).

The uneven development of inclusion in the four countries is also predetermined by the differences in economic situation, and the funding for social projects and programs. According to international rankings, in particular the Childhood Index (Save the Children, 2021), in general, the development of social policies for children varies in four countries of Central Asia. The best situation is in Kazakhstan (49th place in the world) and Uzbekistan (62nd place in the world), and worse in Kyrgyzstan (92nd place in the world) and especially in Tajikistan (105th place in the world).

Poverty as a factor of school exclusion is observed in Kyrgyzstan and Tajikistan, the countries with the lowest GDP per capita in Central Asia (Table 1). The insufficient financing of social policy programs in the context of secondary school education hinders the development of infrastructure as well as methodological complex and advanced training programs for teachers to promote inclusion. In Kyrgyzstan and Tajikistan, inclusive education is developing unevenly in urban and rural areas, in different regions of the country, there is discrimination against girls with disabilities, who are much less enrolled in school than boys (Akkozhoeva, 2020; UNICEF, 2016).

³ Law of the Republic of Kazakhstan “On Ratification of Convention on the Rights of Persons with Disabilities”, https://online.zakon.kz/Document/?doc_id=31668181&pos=2;106#pos=2;106.

⁴ Law of the Kyrgyz Republic “On Ratification of the UN Convention on the Rights of Persons with Disabilities, adopted by the General Assembly of the United Nations on December 13, 2006 and signed on September 21, 2011”// <https://cbd.minjust.gov.kg/act/view/ru-ru/111885?cl=ru-ru>.

⁵ Law of the Republic of Uzbekistan “On Ratification of the Convention on the Rights of Persons with Disabilities”, <https://www.lex.uz/uz/docs/5447430>. <https://www.lex.uz/uz/docs/5447430>.

⁶ The Number of Recipients of Social Benefits for Disability is Growing// National Statistical Committee of the Kyrgyz Republic, <https://stat.kg/en/news/rastyot-chislo-poluchatelej-socialnyh-vyplat-po-invalidnosti/>.

⁷ How to survive for children with disabilities in Tajikistan? BBC, <https://www.bbc.com/russian/features-40527072>.

⁸ Inclusive education in the Republic of Kazakhstan: what is provided for in the new draft law, <https://strategy2050.kz/ru/news/inklyuzivnoe-obrazovanie-v-rk-cto-predusmotreno-v-novom-zakonoproekte/>.

⁹ Concept for the development of inclusive education in the public education system in 2020–2025//Lex-UZ online: <https://lex.uz/ru/docs/5044745#5050709>.

¹⁰ Persons with disabilities, Important figures and facts// National Statistical Committee of the Kyrgyz Republic: <https://www.stat.kg/en/news/lica-s-ogranichenymi-vozmozhnostyami-zdorovya-vazhnye-cifry-i-fakty/>.

Table 1
The sample description.

Country	N, teachers
Kazakhstan	382
Kyrgyzstan	283
Tajikistan	81
Uzbekistan	123

Stigmatization and discrimination of children with disabilities as well as negative attitudes of the population towards school inclusion are important obstacles for inclusive education in Central Asia. Analytical international reports show that this problem is most typical for Uzbekistan and Tajikistan, that is, those countries of Central Asia that have just ratified the Convention or did not ratify it at all by the beginning of 2023. Thus, according to public opinion polls in Uzbekistan, 70 % of the population believes that children with disabilities should study in specialized correctional schools (United Nations, 2019). Stigma and discrimination against children with disabilities in Tajikistan are sometimes observed at the level of their own families, who prefer specialized institutions to mainstream education schools or refuse that children with special educational needs study at school at all (Lapham, 2018).

The level of professional competence of teachers in inclusive education is also unevenly represented across countries. Teachers in four countries of Central Asia must take a refresher course at least every five years, Kazakhstan also has a plan to change the frequency of training courses to once in three years.¹¹ However, training programs on inclusive education are not mandatory for all teachers in Central Asia. Some success in school inclusion has been achieved in Kazakhstan, where more than 7/10 school teachers have completed specialized training on inclusion organized by the country's leading universities, state organizations, and NGOs. Training and programs are aimed at adopting inclusive methods of education, organizing individual learning in a group, and assessing the individual needs of different groups of children (UNESCO, 2021). At the same time, as noted by Allan and Omarova (2021), the segregation of children with disabilities is still taking place. Also, inclusion is perceived by teachers more as a problem, than as a norm of life (Allan & Omarova, 2021). In other countries of Central Asia, educational programs for inclusion are hampered for various reasons. In Uzbekistan, there are lack of professional standards for teachers, and the topic of inclusion is not included in the teacher's professional training programs (United Nations, 2019). Also, the curricula of pedagogical higher educational institutions in Uzbekistan do not include courses on the methodology of inclusive education and students do not practice in educational institutions where children with special educational needs study.¹² As for Tajikistan, uniform educational standards for the whole country have not been developed yet (UNESCO, 2021). Also, by now there are no special state programs for integration of children with special needs and disability in primary, secondary, and higher education. The poor qualification level, as well as the low motivation of teachers, are articulated as challenges in official governmental documents of Tajikistan.¹³ As a response to challenges, grassroots parent associations are being formed in the countries to promote the policy and practice of school inclusion. Such associations participate in filling a

positive agenda around inclusive education, contribute to the methodological training of teachers, and the creation of inclusive schools and classes (Witsel and Kodirov, 2013; Allan & Omarova, 2021).

3. Methodology

3.1. Research approach and design

The empirical basis of the research was the survey "Attitude towards inclusive education in Russia and post-socialist countries"¹⁴ which was conducted by XXX in partnership with NGOs and representatives of the academic community in ten post-Soviet countries in 2020–2021. The survey comprises a wide range of data on different issues of conditions and factors of inclusive education of children with special educational needs, including attitudes of teachers towards inclusive education of children with SEND. Developing this survey, we have adopted the conceptualization of social attitudes (de Boer, 2011) toward teachers in cross-cultural and comparative perspectives. In this quantitative study using a seven-point Likert scale survey questionnaire, the attitudes of eight hundred and sixty-nine (869) teachers from Central Asian countries i.e. Kazakhstan, Kyrgyzstan, Tajikistan, and Uzbekistan were surveyed. The survey samples were as follows:

3.2. Data collection procedure and analysis

The data was collected using a non-probability, convenience sample. This usage sampling method is a limitation of our study, as it may hinder the generalizability of its results. The obtained results on teachers' attitudes toward school inclusion require further verification in a larger-scale study using a probability sample. However, it will be very difficult to implement such a sample due to the closed nature of societies in the countries under consideration and the difficulty of accessing relevant information. The online questionnaire was posted in special chats for parents and teachers in countries of Central Asia and distributed by our partners from local NGOs to potential respondents as well. The questionnaire in Kyrgyzstan was distributed in Russian, Kazakhstan, Uzbekistan, and Tajikistan in two languages – national and Russian. That decision was made based on the results of consultations with colleagues and partners in the countries of Central Asia. In terms of the order of questions and their content, the questionnaires in Russian and in the national language were identical, and the semantic identity of translations into local languages was strictly controlled by experts from countries of Central Asia. In the first stage, using the principal component analysis, the structure of teachers' attitudes towards inclusive education was defined. Here each obtained component represents a separate attitude. Therefore factor scores which were calculated with a regression method (sum of values of each variable multiplied on factor loading with corresponding component) represent the values of each attitude. They are like cumulative indexes, measured on an interval scale, which means the rate of the severity of each attitude. The greater 0 is the factor score, the higher the rate of severity of each attitude. The less than 0 is the factor score, the lower is the rate of severity of each attitude. Then, using a one-way analysis of variance, the severity of identified teachers' attitudes was compared. After that, using hierarchical clustering, a classification of teachers according to the degree of severity of the identified attitudes was done. The choice of such a complex technique for identifying attitudes towards inclusion is determined by the following circumstances. Attitudes themselves are a multidimensional latent trait that has a large number of components with their own sets of indicators. To identify the latent structure of the teachers' attitudes towards inclusion, the principal component method

¹¹ Refresher programs for teachers in the Republic of Kazakhstan: updating of curricula and professional development//Kazinform, available at https://www.inform.kz/ru/povysheniye-kvalifikacii-pedagogov-v-rk-obnovlenie-programm-i-professional-NOE-razvitiye_a3918081.

¹² Concept for the development of inclusive education in the public education system in 2020–2025// Lex-UZ online. available at: <https://lex.uz/ru/docs/5044745#5050709>.

¹³ National Education Development Strategy of the Republic Tajikistan until 2020, available at: https://www.ilo.org/dyn/youthpol/en/equest.fileutils.dochandle?p_uploaded_file_id=511.

¹⁴ The research was approved by the HSE: Meeting Minutes of the HSE Committee on Interuniversity Surveys and Ethical Assessment of Empirical Research May 23, 2023 No. 107.

was used. The result of this analysis is a set of components showing the degree of expression of the attitude. Following [Iskakova, Prisiazhniuk, and Zangieva \(2023\)](#) we assume that teachers may express several attitudes to varying degrees as well as groups of teachers could be char-

4. Findings and discussion

Firstly, we will describe our sample of teachers from 4 observed countries.

	Four countries %	Kazakhstan	Kyrgyzstan	Tajikistan	Uzbekistan
Gender					
Male	11 %	6.8%	1.1%	36.8%	11.3%
Female	89 %	93.2%	98.9%	63.2%	88.7%
Age					
<=30	20.5%	20.8%	21.6%	25 %	14.5%
31–50	65 %	69.2%	48.9%	66.1%	72.6%
51+	15.5%	10 %	29.5%	8.9%	12.9%
Experience in inclusion					
Familiar with inclusion	83.8%	80.7%	86.6%	81.0%	89.4%
Not familiar with inclusion	16.2%	19.3%	13.4%	19.0%	10.6%
Education					
Secondary specialized education	9.9%	13.1%	12.8%	1.7%	5.4%
Incomplete higher education	3.7%	3.8%	6.4%	1.7%	1.4%
Higher: Bachelor(BA)/Specialist	56.8%	63.1%	47.7%	48.3%	63.5%
Higher: Master (MA)	26.6%	16.9%	32.1%	45.0%	24.3%
Candidate of science (PhD)	1.2%	0.6%	0 %	1.7%	4.1%
Doctor of Science (PhD)	1.7%	2.5%	0.9	1.7%	1.4%
Work experience in the field of education					
<=5	25.1%	31.3%	18.5%	14.0%	29.9%
6–10	22.9%	27.2%	13.0%	33.3%	17.9%
11–20	28.4%	25.2%	28.3%	36.8%	28.4%
21–30	15.4%	10.9%	20.7%	14.0	19.4%
31+	8.3%	5.4%	19.6%	1.8%	4.5%
Location and population					
«City-Millionaire» (1 000 000+)	17.4%	25.4%	13.8%	3.6%	19.0%
Very large city (500 000–1 000 000)	18.6%	18.9%	12.6%	38.2%	8.6%
Large city (100 000–500 000)	22.0%	24.6%	20.7%	3.6%	36.2%
Medium or small city (<100 000)	30.1%	23.0%	43.7%	18.2%	36.2%
Village, aul	11.8%	8.2%	9.2%	36.4%	
Questionnaire language					
Russian	26.3%	16.2%	100 %	6.2%	70.7%
Kazakh	54.6%	83.8%			
Tajik	13.0%			93.8%	
Uzbek	6.1%				29.3%

acterized by similar expressions of the same attitudes. However, since such characteristics are unknown, exploration strategy and cluster analysis were used. At the final stage, the representation of selected groups of teachers among countries of Central Asia was compared.

3.3. Ethical considerations

All respondents were adult professionals discussing their attitudes, professional expertise, and work. Before filling out the questionnaire, each respondent was provided with information sheets about the research, its goals, and expected results. Measures were undertaken to protect sensitive information from the fieldwork. The research was conducted on the principle of anonymity and confidentiality, i.e, the names, positions, and other personal information of respondents wasn't collected. The participation of respondents in the research was completely voluntary. Declining to participate didn't involve any penalty or loss of benefits for them. Every respondent could discontinue participation at any time without any penalty. Only the research team from our laboratory had access to the study data.

The sample included teachers with different socio-demographic and professional characteristics. The typical respondents in all cases except Tajikistan lived in urban areas: no more than 10 % of teachers from rural areas in Kazakhstan, Kyrgyzstan, and Uzbekistan were surveyed, while in the Tajik sample, such teachers were more than 36 %. In terms of location, the Tajik sample of teachers was the most binary, because it consisted of respondents from both large cities and rural areas. Also, a distinctive feature of the Tajik sample was the significantly greater representation of male teachers in comparison to other countries. In other respects, the samples of teachers from countries of Central Asia were quite comparable.

Indicators used in the survey were classified as affective or behavioral components of teachers' attitudes. As for the affective component, all indicators reflecting the attitude of teachers toward certain aspects of inclusive education were selected: supporting the idea of inclusion in general (Q8), assessing the benefits of inclusion, and the readiness of the school to work with children with SEND (Q16), assessing the consequences of inclusive education for children with SEND (Q17). The behavioral component of teachers' attitudes was determined through

specific actions related to inclusive education. This aspect was addressed in only one question concerning the assessment of the teachers' readiness to adapt to the needs of students with SEND and to explain the intention of inclusive education to other actors of the educational process (Q18).

Based on these indicators, the set of 29 observed variables was formed representing the 7-item scale of agreement with the statements. The results of the principal component analysis show that the latent components reflecting a certain attitude towards inclusive education lie behind the observed indicators. This is evidenced by the rejection of the statistical hypothesis according to Bartlett's test of sphericity ($\chi^2 = 4436, 503$; $p = 0.000$) and the value of the Kaiser-Meir-Olkin statistics, which is quite close to 1 ($KMO = 0.888$).

As a result of several models' comparison, a six-component model with Promax oblique rotation was chosen for data analysis, because it is optimal in terms of quality (the total share of the explained variance of the observed indicators is 57.9%, the generality is in the range of 0.462–0.714) and the logic of interpretation. The matrix of factor loadings was obtained as follows in [Appendix 1](#):

According to the totality of observed indicators that most closely correlate with each factor (the matrix does not display factor loads less than 0.4), the resulting 6 components, i.e., teachers' attitudes towards inclusive education were named (interpreted based on indicator with maximum factor loading) as follows:

(1) Inclusion is useful, but it requires special arrangements and resources (explains 22.3 % of observed indicators). For this attitude, the main (with maximum factor loadings) indicators are based on the agreement with the following judgments:

- Children with SEND could be at school together with children without SEND if the school has teachers with competencies (0.823);
- Children with SEND should be provided with the opportunity to participate in everyday class life to the possible extent (0.767);
- Children with SEND can study in a school with children without SEND if the school has a resource class and teachers (0.766).

(2) Anxiety of various aspects of inclusion (explains 16.5 % of observed indicators). Here are the main indicators based on the agreement with the following judgments:

- I am concerned that my workload will increase if there are students with SEND in my class (0.782);
- I'm afraid that students with SEND will not be accepted by the rest of the class (0.761);
- I am concerned that inclusion may affect the performance of students without SEND negatively (0.761).

(3) Readiness of the school's infrastructure and teachers for inclusive education (explains 7.4 % of the observed indicators) is operationalized by the indicators below:

- Mainstream teachers have sufficient training to teach children with SEND (0.698);
- The equipment of your school allows you to teach a child with SEND without any problems (0.691);

- The personnel composition (availability of tutors/other specialists) of the school allows to teach children with SEND easily (0.653).

(4) Personal readiness of the teacher for inclusive education (explains 4.2 % of the observed indicators). The main indicators are as follows:

- I am confident that I have the knowledge and skills necessary to teach students with SEND (0.794);
- I can cope with the stress caused by students with SEND in my class (0.675);
- I am confident that all students, including children with special needs, are paid equal attention (0.646).

(5) Disapproval of school inclusion (explains 3.9 % of observed indicators). Here are the main indicators:

- Studying in an inclusive school is a challenge for students with SEND (0.682);
- I believe that the school has the legal right to refuse education to a child with SEND (0.670);
- If parents of children without SEND write a statement to higher authorities, the school will be required by law to transfer children with SEND away from the school (0.642).

(6) Inclusion is a necessity and the right, but also a challenge (explains 3.6 % of the observed indicators). The main indicators based on the agreement with the judgments are as follows:

- Most likely, a child with special needs will have behavior problems in a general education class (0.681);
- Parents' free choice of school for their children, including children with SEND, is the realization of children's legal right to education (0.601);
- It is most likely that an inclusive school/class will prepare children with SEND for adulthood (0.595).

The values of the obtained components showing the degree of severity of each attitude were calculated by the regression analysis. Then, using a one-way ANOVA, the average severity of the identified attitudes was compared among teachers from the four countries of Central Asia.

The results of a one-way ANOVA show that only the severity of the attitude towards the readiness of school's infrastructure and teachers for inclusive education is the same ($\text{sig.} = 0.228$) among teachers from the four countries. The average values of this attitude are close to 0 in all four cases indicating rather a neutral expression of this attitude among teachers.

The average severity of other attitudes (average factor scores listed in [Table 2](#)) differs in at least two countries at the 99 % confidence interval. Further, to reconfirm the results for the first, fourth, and sixth attitudes Scheffe test was used since the dispersion of the severity of attitudes in the four countries turned out to be different. To refine the results for the second, third, and fifth attitudes, the Games-Howell test was used, since the dispersion of the severity of these attitudes was the same.

Table 2

The average severity of attitudes towards inclusive education among teachers from countries of Central Asia (average factor scores).

Country	Inclusion is useful, but it requires special arrangements and resources	Anxiety of various aspects of inclusion	Readiness of the school's infrastructure and teachers for inclusive education	Personal readiness of the teacher for inclusive education	Disapproval of school inclusion	Inclusion is necessity and the right, but also a challenge
Kazakhstan	-.0712751	-.0190514	.1051923	.2326836	.1916521	.1258199
Kyrgyzstan	.3054932	.1687792	-.1514674	-.1525122	-.1874891	.0311318
Tajikistan	-.5027302	-.5263797	.0040246	-.2701831	-.2141159	-.6412343
Uzbekistan	.1182473	.2140550	-.0250025	-.0980261	.0012840	.1723496

Table 3

Classification of teachers according to the severity of attitudes towards inclusive education (average values of factor scores).

Cluster	Inclusion is useful, but it requires special arrangements and resources	Anxiety of various aspects of inclusion	Readiness of the school's infrastructure and teachers for inclusive education	Personal readiness of the teacher for inclusive education	Disapproval of school inclusion	Inclusion is a necessity and the right, but also a challenge	Number of teachers
1	-0.3309525	0.0320414	0.2866385	-0.1491852	0.4182802	-0.4504521	113
2	0.7078492	0.1725715	0.0280573	-0.3620386	-0.757421	0.2542694	88
3	-0.4659553	1.0463039	-1.3398167	-0.22599	0.5920449	0.8260452	67
4	0.6269965	0.9854061	0.5895155	1.1673451	-0.4404746	0.1632662	69
5	0.8245362	1.1788759	1.0788406	0.907477	1.496724	0.968219	25
6	-1.3079602	0.9806101	-0.2895241	-0.8948883	-0.6205677	-1.4156931	44

The attitude “*Inclusion is useful, but it requires special arrangements and resources*” is the least typical for teachers from Tajikistan, and to a greater extent this attitude is the most illustrative for the teachers from Kyrgyzstan. The attitude of “*Anxiety of various aspects of inclusion*” is equally (from neutral to a weakly positive extent) expressed in Kazakhstan, Kyrgyzstan, and Uzbekistan and not typical for Tajikistan. The attitude of “*Personal readiness for inclusive education*” is more appreciated in Kazakhstan than in other countries of Central Asia. The attitude “*Disapproval of school inclusion*” is more prevalent in Kazakhstan than in Kyrgyzstan and Tajikistan (although the severity is still extremely low). This attitude is significantly unusual for teachers from Tajikistan. The attitude “*Inclusion is a necessity and the right, but also a challenge*” is statistically expressed neutrally in Kazakhstan, Kyrgyzstan, and Uzbekistan, and it does not apply to teachers from Tajikistan. According to this data, the teachers from Tajikistan are not characterized by all the indicated attitudes, probably because of the lack of experience and practice in inclusive education as it has not ratified the Convention (Baldiga et al., 2017).

Further, we have noticed that teachers in each country are not characterized by any one dominant attitude, but rather by a combination of expressed or unexpressed attitudes. To classify the teachers according to the expressiveness (severity) of the selected attitudes the cluster analysis was used. The model with 6 clusters was chosen as the final one, obtained by Ward's Hierarchical Clustering Method (Table 3).

The largest group of teachers (named *Passive Disapprovers of Inclusive Education*) based on the severity of attitudes towards inclusive education can be interpreted as teachers who believe that children with SEND may not study in a mainstream school and, to a lesser extent, by the opinion that the school and other teachers are ready for inclusive education. The remaining 4 attitudes do not apply to this group. In the second group (named *Passive Approvers of Inclusive Education*) we interpret a group of

teachers who believe that inclusion is useful, although requires additional resources and teacher's readiness. The third group represents teachers (named *Argued Disapprovers of Inclusive Education*) who are concerned about various aspects of inclusion and believe that while inclusion for children with SEND is the right, it is also a challenge. Moreover, this group of respondents thinks that the infrastructure and teachers should be ready for inclusive education. We interpret the fourth group as teachers (named *Active Approvers of Inclusive Education*) who have positive attitudes towards inclusive education. They exhibit a high personal readiness for inclusive education, they are confident about the readiness of the school and teachers for school inclusion. They tend not concerned about its various aspects and believe inclusive education is a right of children with SEND. The fifth group of teachers (named *Teachers at the Crossroads*) tended to be the most controversial. It is characterized by the highest degree of denial of the rights of children with SEND to inclusive education and a high level of concern about various aspects of school inclusion. Conversely, tend to believe that the school and other teachers are ready for inclusive education and have the highest personal readiness for school inclusion education. Also, they share the position that inclusion is useful, although the availability of resources could be a challenge. We interpret the sixth group of teachers (named as *Teachers having no specific attitudes*) as representing those who do not clearly show any inclination toward all the selected attitudes.

Differences in the representation of each group of teachers among countries were compared using a cross-table analysis and a Chi-square test. The analysis shows a relationship between clusters and countries (the chi-square test is 96.960; $p < 0.001$) (Table 4).

Passive disapproval is more prevalent among teachers in Kazakhstan, and to a much lesser extent for teachers from the other three countries. Passive approval of inclusion was shown mostly by teachers from Kyrgyzstan, and the least from teachers from Tajikistan. The argued

Table 4

Representation of selected groups of teachers among countries of Central Asia (% of the column).

Country	Passive Disapprovers	Passive Approvers	Argued Disapprovers	Active Approvers	Teachers at the Crossroads	Teachers have no specific attitudes
Kazakhstan	48.7%	27.3%	47.8%	53.6%	44.0%	20.5%
Kyrgyzstan	19.5%	44.3%	26.9%	20.3%	20.0%	18.2%
Tajikistan	18.6%	8.0%	0	7.2%	8.0%	52.3%
Uzbekistan	13.3%	20.5%	25.4%	18.8%	28.0%	9.1%

disapproval of inclusion is completely rare for teachers from Tajikistan and is more common among teachers in Kazakhstan, and although lesser, it is equal for teachers from Kyrgyzstan and Uzbekistan. Active approval of inclusion is dominant among Kazakhstani teachers and much less expressed by teachers from Tajikistan. The teachers at the crossroads were mostly from Kazakhstan and least from Tajikistan. The majority of teachers with no specific attitudes were from Tajikistan and fewer were from Uzbekistan.

The data shows that teachers from Kazakhstan were most represented in $\frac{2}{3}$ groups of teachers. This could be attributed to the fact that teachers from Kazakhstan dominate the sample (Table 1). Furthermore, it may also indicate that Kazakhstan has the longest history of school inclusion among other countries in Central Asia. It is the first country in the region which ratify the Convention and adopted the legislation for inclusive education. It goes to show that teachers in Kazakhstan have already developed very different, both positive and negative, experiences that led to ambivalent attitudes toward school inclusion. It also corresponds with the ambivalent results of the previous surveys, described in our literature review: inclusive training programs, more common in Kazakhstan than among other countries of Central Asia, which may have affected teachers attitudes towards school inclusion both positively (Agaveyan et al., 2020) and negatively (Gyimah et al., 2010).

Teachers from Kyrgyzstan are mostly represented in the group of Passive Approvers. Firstly, the positive attitude towards school inclusion in Kyrgyzstan is influenced by experienced teachers who constituted the majority of respondents – 68 % of teachers from Kyrgyzstan had work experience of more than 10 years (Unianu, 2012). Secondly, Kyrgyzstan had already ratified the Convention in 2019 and begun to implement school inclusion, including training programs for teachers. This could affect their attitudes to be more positive. At the same time, teachers from Kyrgyzstan regard insufficient financing of schooling (Akkozhoeva, 2020) as a barrier to the success of inclusion. They support the idea that inclusion requires additional resources and training.

Teachers from Uzbekistan show more ambivalence and even negative attitudes towards school inclusion. In comparison with other countries of Central Asia, the less experienced group of teachers was from Uzbekistan, 30 % had work experience of less than five years in the field of education. This is consistent with the findings of Unianu (2012) who found that low work experience in inclusion may predispose teachers toward negative attitudes. Furthermore, negative attitudes may be connected to the institutional processes of developing school inclusion in Uzbekistan, such as the time of ratification of the Convention (2021, before the survey was conducted), the prevalence of medical model of disability in the country, and negative attitudes towards inclusive education among population at large (United Nations, 2019). The ratification of Convention requires further school inclusion be promoted in legislation and practices as this may affect teacher's attitudes towards inclusion in a positive way. It also requires the creation of support institutions and professional learning communities that may need time to alter teachers' attitudes positively (Gyimah et al., 2010).

The prospects for inclusive education in Tajikistan are complicated. There is a challenge to the legislation, financial resources for inclusive infrastructure, and training programs implementation. Practically, a high proportion of teachers from Tajikistan show no attitudes towards inclusive education or no specific experience with it. This is indicative of the indifferent attitudes of teachers from Tajikistan. In this study, we noted that the proportion of male teachers and those residing in rural and small towns was the biggest in Tajikistan in comparison to samples from the other three countries. Previous studies have associated such with more negative attitudes towards school inclusion (Gyimah et al., 2010; Saloviita, 2020). Be that as it may, teachers from Tajikistan are experienced, have a high level of education, and long work service in the field of education, factors which are mostly associated with positive attitudes towards inclusive education in previous studies (Bandyopadhyay & Dhara, 2021, Unianu, 2012).

While reflecting on the study, we recognize that samples of teachers from Kazakhstan, Kyrgyzstan, and Uzbekistan were more or less comparable in terms of size, but the sample of teachers from Tajikistan was significantly smaller. This we think constituted the major limitation of the study.

5. Conclusion

The study explored teacher's attitudes toward inclusive education and SEND in Central Asian countries. The analysis from the study shows, however, a noticeable shift from the segregation strategy embedded in defectology, as a special education approach that was practiced in the former Soviet Union to inclusion which embraces diversity, equality, and equity for all students (Makoelle, 2020).

According to Baldiga, Gattuso, Baker, Gruber, and Deaton (2017), the ratification of the Convention usually becomes a driver for school inclusion and causes more positive attitudes of various schooling actors, including teachers. The results of the analysis suggest that relationships between the ratification of the Convention and teachers' attitudes could be more complicated. The Tajikistan where the Convention is not ratified yet have no certain attitudes towards inclusive education. Attitudes of teachers towards school inclusion in Uzbekistan, where the Convention was ratified only in 2021, tend to be ambivalent and even negative, which could be explained by unformed institutions of inclusive education. Kyrgyzstan ratified the Convention in 2019 which could cause more positive teachers' attitudes towards inclusive education. But Kyrgyz teachers' position is rather moderate, they tend to suggest that inclusion requires additional resources while teachers themselves are rather unprepared for it. Following the logic of UNICEF, the teachers from Kazakhstan, which is the pioneer of school inclusion in the region, should be the most positive towards school inclusion in comparison with the respondents from other countries of Central Asia. Practically, these teachers compile different attitudes towards inclusion, both positive and negative, which reflects their everyday life experience of schooling practices. These findings are consistent with the study by Boyle, Anderson, and Allen (2020) who postulate that although the attitudes of teachers could change in a positive direction, a policy framework that guides the implementation could have far-reaching implications for teachers' willingness to embrace inclusion. It is also evident in the study that resources provision for support of teachers may influence their attitudes positively or negatively (Gitschthaler, Kast, Corazza, & Schwab, 2021). Furthermore, factors such as teacher perceptions about SEND (Lindner, Schwab, Emara, & Avramidis, 2023), their workload, experience (Galović et al. (2014)), level of education (Bandyopadhyay & Dhara, 2021), preparedness for inclusion, including training programs (Koliqi & Zabeli, 2022), beliefs about inclusion (Makoelle et al., 2023), and the infrastructure seems to be critical in teachers attitudes towards inclusive education in other contexts (Makoelle et al., 2023). These factors also emerged in this study as prominently affecting the attitudes of teachers in Central Asian countries.

While the evidence from the research contributes to understanding factors of school inclusion and the role of teachers' attitudes in developing inclusive education in Central Asia, and such contributions could be relevant for further academic discussion as well as for policy-makers, it must be born in mind that in the Central Asian countries school inclusion as policy and practice is a fairly new concept and the only way to make it possible is to ensure teacher readiness, their capacity building, resource provision and the creation of enabling inclusive environment in schools.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgements

The article was prepared within the framework of the Basic Research Program at HSE University.

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